I590 Topics in Informatics: Introduction to Genomics

Project Proposal

Write a proposal for one of the following four topics to accomplish its goal. You may use the techniques you learned throughout the courses. Once you choose your topic, visit the project topic selection webpage and make sure your selection is secured. The last person will have to make up his/her own topic. You may volunteer for your own topic if you want to.

Webpage

http://biokdd.informatics.indiana.edu/cgi-bin/juhur/I590_Genomics/Project.cgi

Requirements

4 pages or more.

Submit an electronic version of proposal to Haixu Tang (hatang@indiana.edu) and Junguk Hur (juhur@indiana.edu).

No specific format but make sure your proposal is properly sectioned.

Figures and tables are welcomed.

Properly cite your references.

Time-line

Topic distribution on	**: March 29th (Tue)**
Topic selection by	**: April 5th (The) –** First come first served basis
Proposal submission by	**: May 2nd (Mon)**

Evaluation

Novelty of the idea

Properness of the used technique

How well addressed the research problem

Overall presentation

Topic1. Cancer

Cancer is any malignant growth or tumor that can be caused by abnormal and uncontrolled cell division. It may spread to other parts of the body through the lymphatic system or the blood stream. Even though the causes of cancers are quite various, they are basically grouped into two categories. The first group lacks ability to properly inhibit cell cycle propagation and the second group has too high level of stimulation to continue cell cycle. In many cases, cancers can be caused initially by the malfunction of single gene. Suppose you are studying one type of cancer and want to find out the cause. Propose how you can find the cause of this cancer of your interest, using the techniques you learned from this course.

Topic2. Expression System

Since we need high level of expressed protein in the industry, we should have a powerful gene expression system. Create your own gene expression system that can express your gene of interest at high efficiency. Address the issues regarding organisms that you want to use, different expressional features (splicing), and most importantly the efficiency of your system.

Topic3. Map Based Cloning

Map-based or Positional cloning is a gene discovery methodology that clones the gene based on its chromosomal location. Here are basic steps of Map based cloning.

1. Identify a **marker** tightly linked to your gene in a "large" mapping population
2. Find a YAC or BAC clone to which the marker probe hybridizes
3. Create new markers from the large-insert clone and determine if they **co-segregate** with your gene
4. If necessary, re-screen the large-insert genomic library for other clones and search for co-segregating markers
5. Identify a candidate gene from large-inset clone whose markers co-segregate with the gene
6. Perform genetic complementation (transformation) to rescue the wild-type phenotype

7. Sequence the gene and determine if the function is known

Refer to “Arabidopsis Map-Based Cloning in the Post-Genome Era, *Plant Physiology*, June 2002, Vol. 129, pp. 440–450” for further detail of map-based cloning.

Suppose you are studying a genetic disease (eg. Sickle cell anemia, Huntington’s disease and etc). Fortunately you found a small population in which the people are highly risky at this disease. Propose a method to identify the gene(s) that relate to this disease based on the high risky population.

Topic4. Cell Cycle Related Gene

The cell cycle is an ordered set of events, culminating in cell growth and division into two daughter cells. Regulation of the cell cycle, how cell division (and thus tissue growth), is very complex. A cell cycle has two distinct phases. During M phase or mitosis phase the replicated chromosomes are segregated into separate nuclei (by mitosis) and the cell splits into two (by cytokinesis). The other, much longer part of the cycle is known as Interphase. This period of continuous cell growth includes S phase, when DNA replication occurs, and two gaps, the G1 and G2 phases, between S phase and M phase. The sequence of cell – cycle events is governed by a cell-cycle control system, which cyclically triggers the essential processes of cell reproduction, such as DNA replication and chromosome segregation. Suppose that you are interested in identifying proteins that are involved in regulating cell cycles. Propose a method that can identify such proteins playing important roles in cell cycle controls.